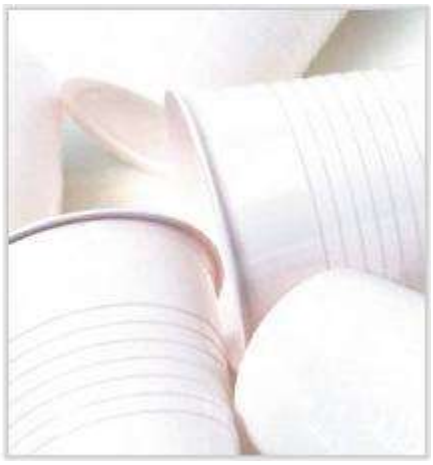
Description

Image





Caption

1. Transparent polystyrene pens. © Carlos Delgado, Kadellar at en.wikipedia - (CC BY-SA 3.0) 2. Close-up of scratches on the material. © Carlos Delgado, Kadellar at en.wikipedia - (CC BY-SA 3.0) 3. Plastic cups. © Chris Lefteri

The material

Polystyrene is an optically clear, cheap, easily molded polymer, familiar as the standard "jewel" CD case. In its simplest form PS is brittle. Its mechanical properties are dramatically improved by blending with polybutadiene, but with a loss of optical transparency. High impact PS (10% polybutadiene) is much stronger even at low temperatures (meaning strength down to -12C). The single largest use of PS is a foam packaging.

Compositional summary

(CH(C6H5)-CH2)n

General properties

Density	1.04e3	-	1.05e3	kg/m^3
Price	* 1.22	-	1.46	GBP/kg
Date first used	1937			

Mechanical properties

Young's modulus	3	-	3.5	GPa
Shear modulus	* 1.08	-	1.25	GPa
Bulk modulus	* 4.76	-	5.55	GPa
Poisson's ratio	* 0.387	-	0.403	
Yield strength (elastic limit)	* 28.7	-	41.4	MPa
Tensile strength	35.9	-	51.7	MPa
Compressive strength	* 82.7	-	89.6	MPa
Elongation	1.2	-	2.5	% strain

Hardness - Vickers		9	-	12	HV
Fatigue strength at 10^7 cycles	*	14.4	-	20.7	MPa
Fracture toughness	*	0.7	-	1.1	MPa.m^0.5
Mechanical loss coefficient (tan delta)	*	0.0122	-	0.0175	

Thermal properties

Glass temperature		89.9	-	99.9	°C
Maximum service temperature		74.9	-	89.9	°C
Minimum service temperature		-75.2	-	-65.2	°C
Thermal conductor or insulator?		Good insulator			
Thermal conductivity		0.12	-	0.14	W/m.°C
Specific heat capacity		1.2e3	-	1.3e3	J/kg.°C
Thermal expansion coefficient		90	-	149	µstrain/°C

Electrical properties

Electrical conductor or insulator?		Good insulator			
Electrical resistivity		1e25	-	1e27	µohm.cm
Dielectric constant (relative permittivity)		2.4	-	2.6	
Dissipation factor (dielectric loss tangent)		1e-4	-	3e-4	
Dielectric strength (dielectric breakdown)		19.7	-	22.6	MV/m

Optical properties

Transparency		Optical Quality			
Refractive index		1.55	-	1.61	

Critical Materials Risk

High critical material risk?		No			

Processability

Castability		1	-	2	
Moldability		4	-	5	
Machinability		3	-	4	
Weldability		5			

Durability: water and aqueous solutions

Water (fresh)		Excellent			
Water (salt)		Excellent			
Soils, acidic (peat)		Excellent			
Soils, alkaline (clay)		Excellent			
Wine		Excellent			

Durability: acids

Acetic acid (10%)	Excellent
Acetic acid (glacial)	Limited use
Citric acid (10%)	Excellent
Hydrochloric acid (10%)	Excellent
Hydrochloric acid (36%)	Limited use
Hydrofluoric acid (40%)	Unacceptable
Nitric acid (10%)	Acceptable
Nitric acid (70%)	Limited use
Phosphoric acid (10%)	Excellent
Phosphoric acid (85%)	Limited use
Sulfuric acid (10%)	Excellent
Sulfuric acid (70%)	Unacceptable

Durability: alkalis

Sodium hydroxide (10%)	Excellent
Sodium hydroxide (60%)	Excellent

Durability: fuels, oils and solvents

Amyl acetate	Unacceptable
Benzene	Unacceptable
Carbon tetrachloride	Excellent
Chloroform	Unacceptable
Crude oil	Limited use
Diesel oil	Limited use
Lubricating oil	Limited use
Paraffin oil (kerosene)	Excellent
Petrol (gasoline)	Limited use
Silicone fluids	Excellent
Toluene	Unacceptable
Turpentine	Unacceptable
Vegetable oils (general)	Limited use
White spirit	Limited use

Durability: alcohols, aldehydes, ketones

Acetaldehyde	Unacceptable
Acetone	Unacceptable
Ethyl alcohol (ethanol)	Excellent
Ethylene glycol	

	Excellent
Formaldehyde (40%)	Limited use
Glycerol	Excellent
Methyl alcohol (methanol)	Acceptable

Durability: halogens and gases

Chlorine gas (dry)	Unacceptable
Fluorine (gas)	Unacceptable
O2 (oxygen gas)	Unacceptable
Sulfur dioxide (gas)	Excellent

Durability: built environments

Industrial atmosphere	Acceptable
Rural atmosphere	Excellent
Marine atmosphere	Excellent
UV radiation (sunlight)	Fair

Durability: flammability

Flammability	Highly flammable

Durability: thermal environments

Tolerance to cryogenic temperatures	Unacceptable
Tolerance up to 150 C (302 F)	Acceptable
Tolerance up to 250 C (482 F)	Unacceptable
Tolerance up to 450 C (842 F)	Unacceptable
Tolerance up to 850 C (1562 F)	Unacceptable
Tolerance above 850 C (1562 F)	Unacceptable

Geo-economic data for principal component

Annual world production, principal component	1.88e7			tonne/yr
Reserves, principal component	2.85e8	-	3.15e8	tonne

Primary material production: energy, climate change and water

Climate change (CO2-eq) (virgin grade production)	* 2.56	-	2.83	kg/kg
Embodied energy (virgin grade production)	* 78.9	-	87	MJ/kg
Water usage	* 144	-	160	l/kg

Material processing: energy

Polymer extrusion, embodied energy	* 7.99	-	8.84	MJ/kg
Polymer molding (injection, blow), embodied energy	* 19	-	21.8	MJ/kg

Machining, coarse, embodied energy	* 19.2	-	21.6	MJ/kg
Machining, fine, embodied energy	* 28.5	-	32.6	MJ/kg
Machining, grinding, embodied energy	* 37.8	-	43.6	MJ/kg

Material processing: climate change

Polymer extrusion, climate change (CO2-eq)	* 0.524	-	0.582	kg/kg
Polymer molding (injection, blow), climate change (CO2-eq)	* 1.29	-	1.49	kg/kg
Machining, coarse, climate change (CO2-eq)	* 1.26	-	1.42	kg/kg
Machining, fine, climate change (CO2-eq)	* 1.87	-	2.14	kg/kg
Machining, grinding, climate change (CO2-eq)	* 2.47	-	2.86	kg/kg

Recycling and end of life

Recycle	✓			
Functional recycle	✗			
Climate change (CO2-eq), recycling	* 0.845	-	0.934	kg/kg
Embodied energy, recycling	* 15	-	16.5	MJ/kg
Recycle fraction in current supply	1			%
Combust for energy recovery	✓			
Heat of combustion (net)	* 40.1	-	42.1	MJ/kg
Combustion CO2	* 3.3	-	3.47	kg/kg
Landfill	✓			
Biodegrade	✗			
Toxicity rating	Non-toxic			
A renewable resource?	✗			

Environmental notes

The flammability of PS foam, and the use of CFC's as blowing agents in the foaming process was, at one time, a cause for concern. New flame retardants allow PS foams to meet current fire safety standards, and CFC blowing agents have been replaced by pentane, CO2 or HFC's which do not have a damaging effect on the ozone layer. PS can be recycled. The large volume of PS foam in packaging, much of it dumped at present, is a cause for concern. The monomer, styrene, is irritating to the eyes and throat, but none survives in the polymer.

Recycle mark



Supporting information

Design guidelines

PS comes in 3 guises: as the simple material ('general purpose PS'); as the high impact variant, blended with polybutadiene; and as polystyrene foam, the most familiar and cheapest of all polymer foams. All are FDA approved for use as food containers and packaging. General purpose PS is easy to mold. Its extreme clarity, ability to be colored, and high refractive index give it a glass-like sparkle, but it is brittle and cracks easily (think of CD cases). It is used when the optical attractiveness and the low cost are sought, and the mechanical loading is light: cosmetic compacts, transparent but disposable glasses, cassettes of all kinds. Medium and high impact polystyrenes trade their optical for their mechanical properties. Medium impact PS, translucent, appears in electrical switch gears and circuit breakers, coat hangers and combs. High impact PS - a blend of PPO and PS, is opaque, but is tough and copes better with low temperatures than most plastics; it is found in interiors of refrigerators and freezers, and in food trays such as those for margarine and yogurt. Other styrene blends, like Kraton, have low tensile strength and higher elongation than SBR or natural rubber. PS can be foamed to a very low density (roughly 1/3 of all polystyrene in foamed). These foams have low thermal conduction and are cheap, and so are used for house insulation, jackets for water boilers, insulation for disposable cups. They crush at loads that do not cause injury to delicate objects (such as TV sets or to the human body), making them good for packaging.

Technical notes

Polystyrene, PS, is - like PE and PP - a member of the polyolefin family of moldable thermoplastics. In place of one of the H-atoms of the polyethylene it has a C6H5 - benzene ring. This makes for a lumpy molecule which does not crystallize, and the resulting material is transparent with a high refractive index. The benzene ring absorbs UV light, exploited in the PS screening of fluorescent lights, but also causing the polymer to discolor in sunlight. All grades of PS have excellent electrical resistance and dielectric strength, exploited in switchgear.

Typical uses

Toys, light diffusers, lenses and mirrors, beakers, cutlery, general household appliances, video/audio cassette cases, electronic housings, refrigerator liners.

Tradenames

Aim, Bapolan, Comalloy, Dylite, Lastirol, NSC, Polystyrol, Styron, Styropor, Vestyron

Links

ProcessUniverse

Producers

Reference